FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 2

B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS (SOLID STATE PHYSICS I)

Module II and Module III (Unit 12): Theory of Solids; Intrinsic and Extrinsic Semiconductors

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	State the basic assumptions of the Drude–Lorentz free electron gas model of metals. State the Wiedemann–Franz law.	**3**
2	Find the relaxation time and mean free path of conduction electrons in copper. For copper, the density of free electrons is $8.5 \times 10^{28}\ \mathrm{m^{-3}}$, the resistivity is $1.69 \times 10^{-8}\ \Omega\,\mathrm{m}$ and the average velocity of the electrons is $1.154 \times 10^{5}\ \mathrm{m\,s^{-1}}$.	**3**
3	Write down the Schrödinger equation for an electron of mass m in a one-dimensional box of length L with infinitely high potential barriers, and write the allowed energy eigenvalues E_n. Why is every energy level doubly degenerate?	**3**
4	Define the Fermi energy. Calculate the Fermi energy of electrons at $0\ \mathrm{K}$ for a metal with electron density $10^{28}\ \mathrm{m^{-3}}$.	**3**
5	Define the density of states. Write the expression for $D(E)$ of a one-dimensional free electron gas and sketch its variation with energy.	**3**
6	Why is the observed electronic specific heat of metals at ordinary temperatures much smaller than the classical value $\frac{3}{2}Nk_B$? Give the expression for C_v obtained from the Sommerfeld theory.	**3**
7	State the Bloch theorem. Write the form of the Bloch function and state the property of $u_{\mathbf{k}}(\mathbf{r})$.	**3**
8	What is the effective mass of an electron in a crystal? Can it be negative? What is its importance?	**3**
9	Distinguish between a metal, an insulator and a semiconductor on the basis of their energy band structure. Give the order of the band gap in each case.	**3**
10	What are intrinsic and extrinsic semiconductors? Distinguish between donor and acceptor impurities, giving one example of each.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Silver (fcc) has an atomic radius of 1.44 Å. Assuming silver to be a monovalent metal, calculate the value of the Fermi energy, the Fermi temperature and the Fermi velocity.	**6**
12	A one-dimensional metallic crystal contains 0.5 electron per Å of its length. Calculate the Fermi energy and the average kinetic energy per electron in the ground state. Why does the Fermi energy depend on the length of the box and the number of electrons?	**6**
13	Derive the pressure versus volume relationship for a free electron gas at $0\ \mathrm{K}$. Hence obtain an expression for the bulk modulus in terms of the average kinetic energy.	**6**
14	How does the classical free electron theory lead to Ohm's law? Explain complete opacity and high lustre of metals on the basis of the Drude–Lorentz theory. State two failures of the theory.	**6**
15	What is meant by the effective mass of an electron? Show that the effective mass of an electron in a crystal is inversely proportional to the second derivative of the E–k curve. Discuss the conditions when the effective mass becomes positive, negative and infinity.	**6**
16	The energy near the valence band edge of a crystal is given by $E = -Ak^2$, where $A = 10^{-39}\ \mathrm{J\,m^2}$. An electron with wave vector $\mathbf{k} = 10^{10}\,\hat{\mathbf{k}}_x\ \mathrm{m^{-1}}$ is removed from an orbital in the completely filled valence band. Determine the effective mass, velocity, momentum and energy of the hole.	**6**
17	Using the effective number of free electrons in a band, explain how the band theory distinguishes between metals, insulators and semiconductors. Why does the conductivity of metals decrease while that of semiconductors increases with rise in temperature?	**6**
18	Explain the concept of a hole in a semiconductor. With the help of covalent bond and energy band diagrams, explain how n-type and p-type semiconductors are formed. State the law of mass action.	**6**

Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Discuss the formation of allowed and forbidden energy bands on the basis of the Kronig–Penney model. Discuss the extreme conditions when energy levels are either discrete or continuous. What is the effect of changing the binding energy of the electron on the energy bands?	**10**
20	Obtain expressions for the Fermi energy, the total energy and the density of states for a free electron gas in one dimension. Show the variation of density of states with energy.	**10**